Vaccine availability and electricity continuity in Nigeria

FASTR Nigeria Round 1 | May to July 2025
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Electricity continuity emerges as a concern across every zone and both programme groups. Nationally, 24.43% of facilities reported continuous electricity in the preceding seven days, compared with 55.91% reporting a primary electricity source. Vaccine availability also varied across zones and programme groups. This profile examines these patterns to identify priorities for follow-up, using published FASTR summaries and a reproducible Stata 19 analysis.

Electricity reliability deserves attention alongside access

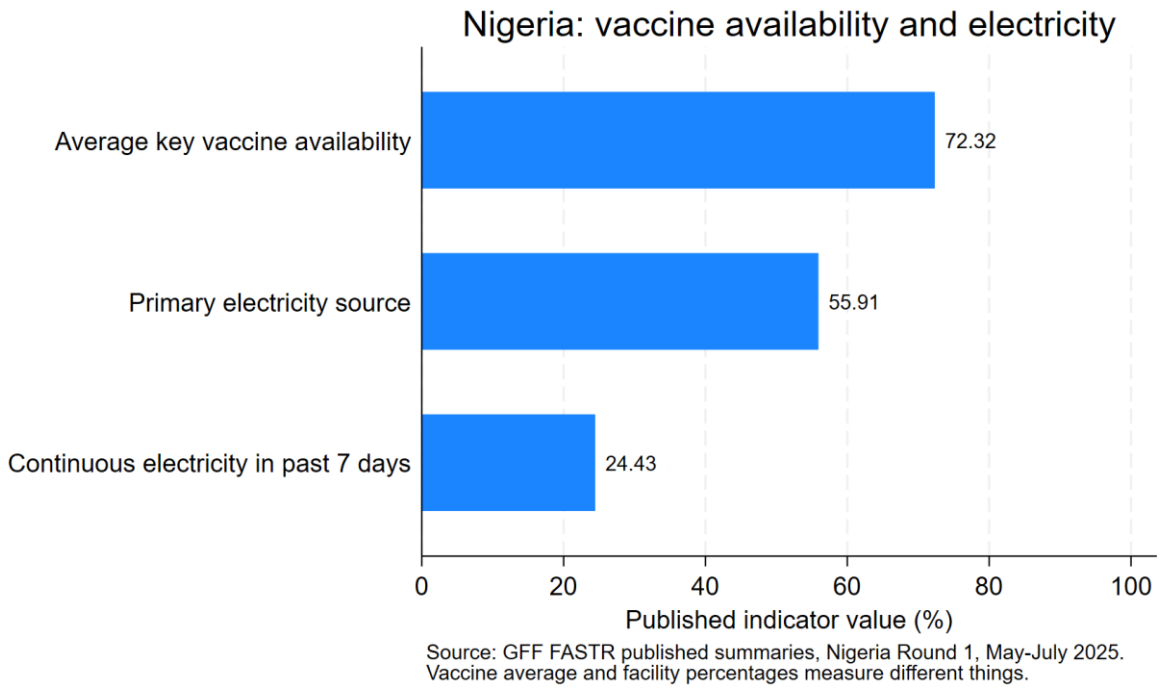


Figure 1. National published summaries. All charts in this report are the original Stata exports.

Table 1. National results from readiness_national_table.csv. Survey sample for each published summary: 3,989.

Indicator	Published value %
Average key vaccine availability	72.32
Primary electricity source	55.91
Continuous electricity in past 7 days	24.43

The national results point to two questions for electricity support: which facilities need access to a source, and where does existing access need to become more reliable? Assessing continuity alongside infrastructure provision would help programme teams distinguish these needs. The vaccine average provides a separate reason to examine the availability of individual vaccines and the supply conditions behind the reported result.

Vaccine availability is the published average across tracer vaccines. Electricity indicators measure the percentage of facilities reporting the stated condition. Definitions, extraction checks and analytical limits are explained on page 4.

Different zones call for different follow up

North-Central had the lowest vaccine availability, while South-South had the lowest electricity continuity. North-West ranked second for vaccine availability but last for a primary electricity source. Reviewing the indicators separately would help tailor follow-up to each zone's reported conditions.

Table 2. All zone results from readiness_zone_table.csv.

Zone	Vaccine average %	Electricity source %	Continuous electricity %	Survey sample
North-Central	68.36	56.56	26.98	720
North-East	69.55	48.17	18.86	518
North-West	75.78	42.56	24.81	869
South-East	81.11	68.87	23.99	610
South-South	74.34	59.91	18.22	572
South-West	70.13	70.38	32.31	700

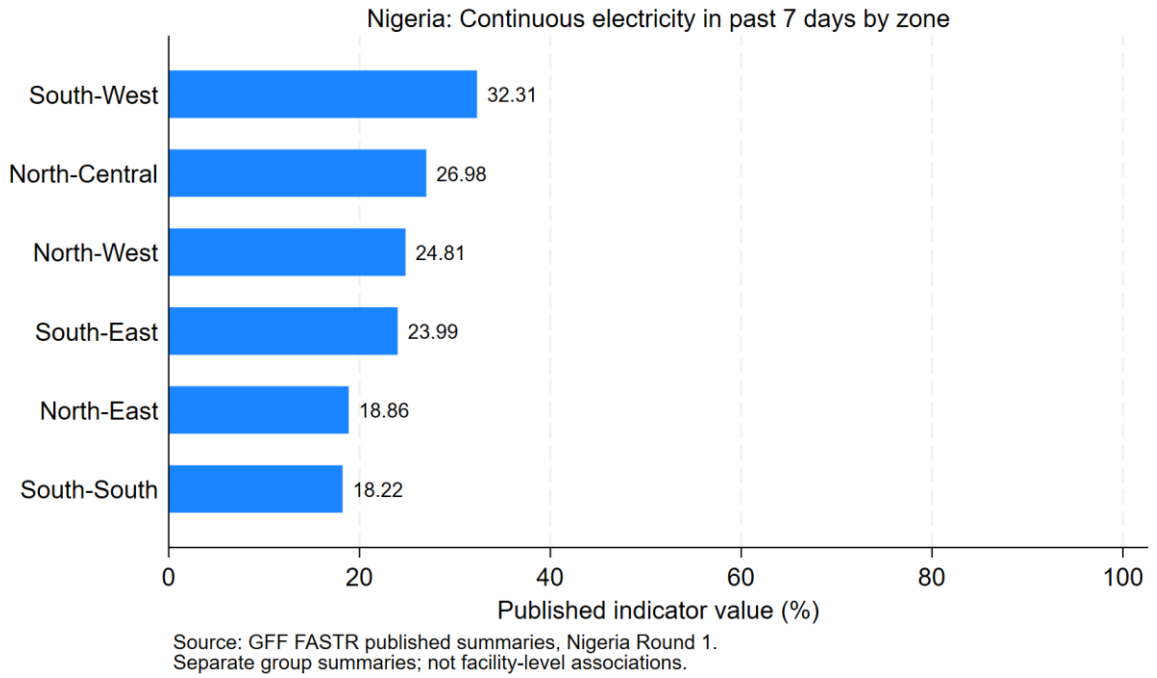


Figure 2. Facilities reporting continuous electricity during the preceding seven days.

Electricity continuity was below one-third in every zone, making reliability a concern across the country as well as a question of regional differences. South-South and North-East merit early attention to continuity. North-Central and North-East merit follow-up on vaccine supply. South-West's higher electricity values and South-East's higher vaccine availability also provide starting points for examining practices that may be useful elsewhere.

Supporting vaccine and electricity-source charts appear in Appendix A. Table 2 retains survey sample sizes for reference.

BHCPF facilities report stronger inputs with room to improve reliability

BHCPF facilities reported higher vaccine availability and electricity values. The largest group difference was in access to a primary electricity source, at 37.99 percentage points. This makes electricity infrastructure an important part of the observed programme-group contrast. Continuous electricity was reported by 36.56% of BHCPF facilities and 18.64% of non-BHCPF facilities, pointing to reliability needs in both groups.

Table 3. All programme results from readiness_programme_table.csv. Survey samples: non-BHCPF 828; BHCPF 3,161.

Indicator	Non-BHCPF %	BHCPF %
Average key vaccine availability	67.43	82.56
Primary electricity source	43.64	81.64
Continuous electricity in past 7 days	18.64	36.56

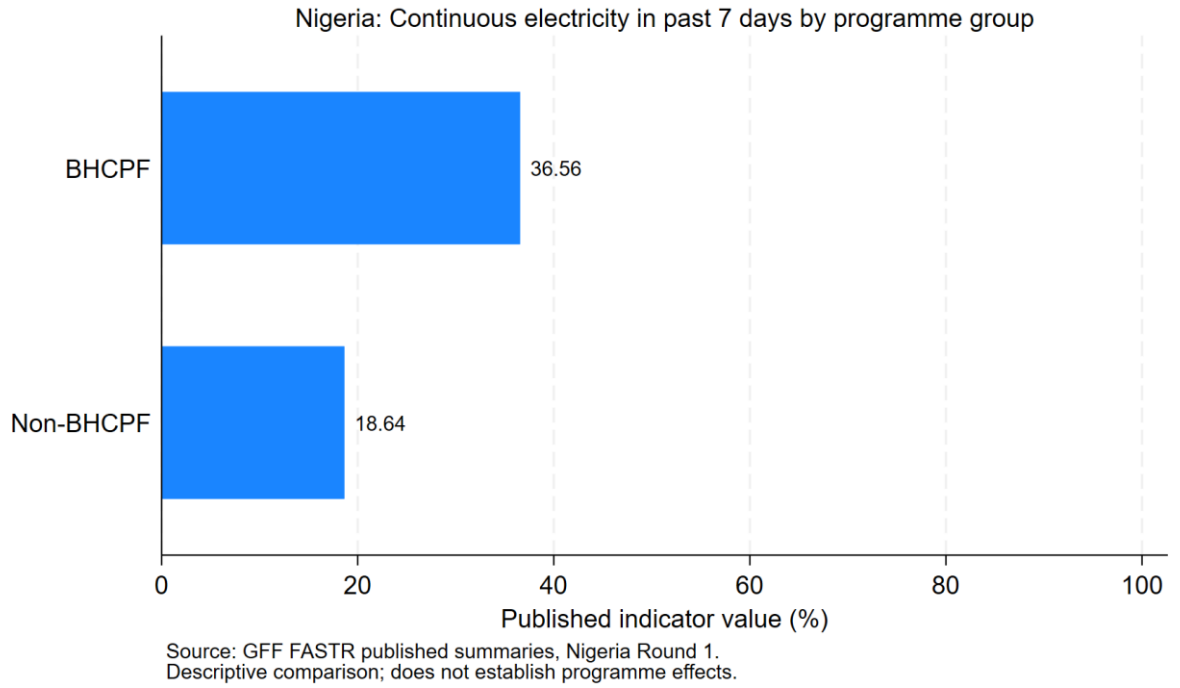


Figure 3. Electricity continuity by Basic Health Care Provision Fund programme group.

For BHCPF facilities, the results draw attention to improving electricity reliability alongside the infrastructure already reported. For non-BHCPF facilities, the lower values across all three indicators warrant examining both infrastructure access and vaccine supply. This distinction can help structure facility reviews around the needs of each group.

Questions for programme review

Programme teams could examine how the higher reported vaccine and electricity values in BHCPF facilities are sustained, and which practices might be useful in facilities with lower values. Within each group, reviews could distinguish needs for equipment, maintenance, operating support and supply follow-up.

Facility-level checks would then establish where interruptions occur, how vaccine availability varies by product, and which local conditions need attention. Together, these checks could inform a more specific support plan for facilities in each group.

Supporting vaccine and electricity-source programme charts appear in Appendix B.

Comparison gaps and follow up

Table 4. All derived gaps from readiness_comparison_gaps.csv. Differences use full-precision source values.

Indicator	Zone minimum %	Zone maximum %	Zone gap pp	BHCPF gap pp
Average key vaccine availability	68.36	81.11	12.75	15.13
Primary electricity source	42.56	70.38	27.83	37.99
Continuous electricity in past 7 days	18.22	32.31	14.09	17.93

pp = percentage points. Zone gap is highest minus lowest; BHCPF gap is BHCPF minus non-BHCPF.

The largest observed differences concern access to an electricity source, across both zones and programme groups. Electricity continuity is lower throughout the profile, including in groups with relatively strong source access.

Methods and reproducibility

The source was GFF FASTR's published summary workbook, FASTR-Data.xlsx, country-round NG_R1, with a May-July 2025 survey period. The do-file selects SUP_05AAA_*, INF_04 and INF_05 by their source codes from the national, region and facility-type sheets. It imports strings, converts numeric values, reshapes subgroup tables and merges indicator metadata using explicit keys. Published national values are retained without reconstructing them from subgroup means.

Assertions check country-round and subgroup headers, survey period, three unique indicator rows per sheet, expected output sizes of 3 national, 18 zone and 6 programme rows, unique keys, complete numeric values, 0-100 ranges and positive integer survey samples. Zone and programme samples each reconcile to 3,989 per indicator. The run completed on 13 October 2026 with no Stata error codes. CSV tables were also checked against their saved DTA datasets.

Limits of the evidence

These are descriptive results from aggregate summaries for one survey round. Survey samples are not confirmed indicator denominators and were not used as weights; confidence intervals and significance tests were not calculated. The separate summaries do not identify overlapping gaps in the same facilities, explain vaccine availability through electricity conditions, or establish BHCPF effects. Geography, facility characteristics and programme selection may contribute to the group differences. Joint zone-by-programme comparisons require further data. Vaccine availability is distinct from vaccination coverage, and general electricity measures do not establish cold-chain function. The work checks published-summary extraction; it does not include raw-response validation, HMIS linkage or service-module validation.

Source: GFF FASTR data portal, https://data.gffportal.org/key-themes/FASTR; source workbook FASTR-Data.xlsx. Results: readiness_national_table.csv, readiness_zone_table.csv, readiness_programme_table.csv and readiness_comparison_gaps.csv. Audit trail: FASTR_Nigeria_expanded.log and readiness_extraction_checks.txt. Code: FASTR_Nigeria_expanded.do. Extraction checks concern published summaries rather than raw facility responses.

Appendix A Supporting zone charts

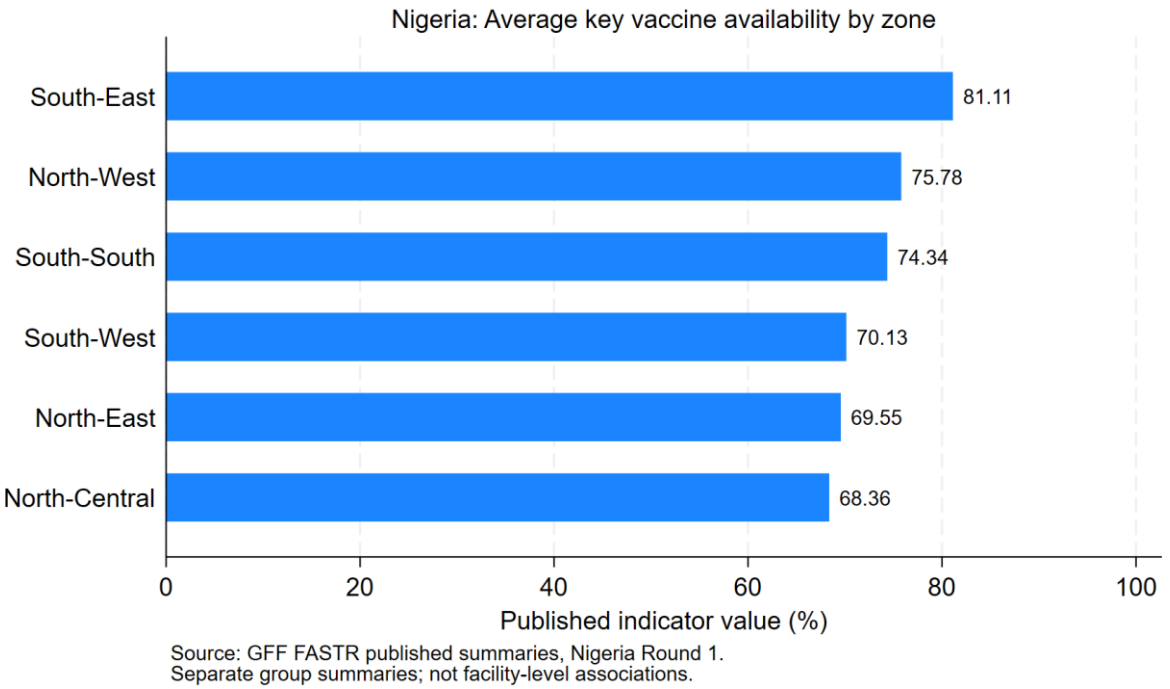


Figure A1. Published average key vaccine availability by zone.

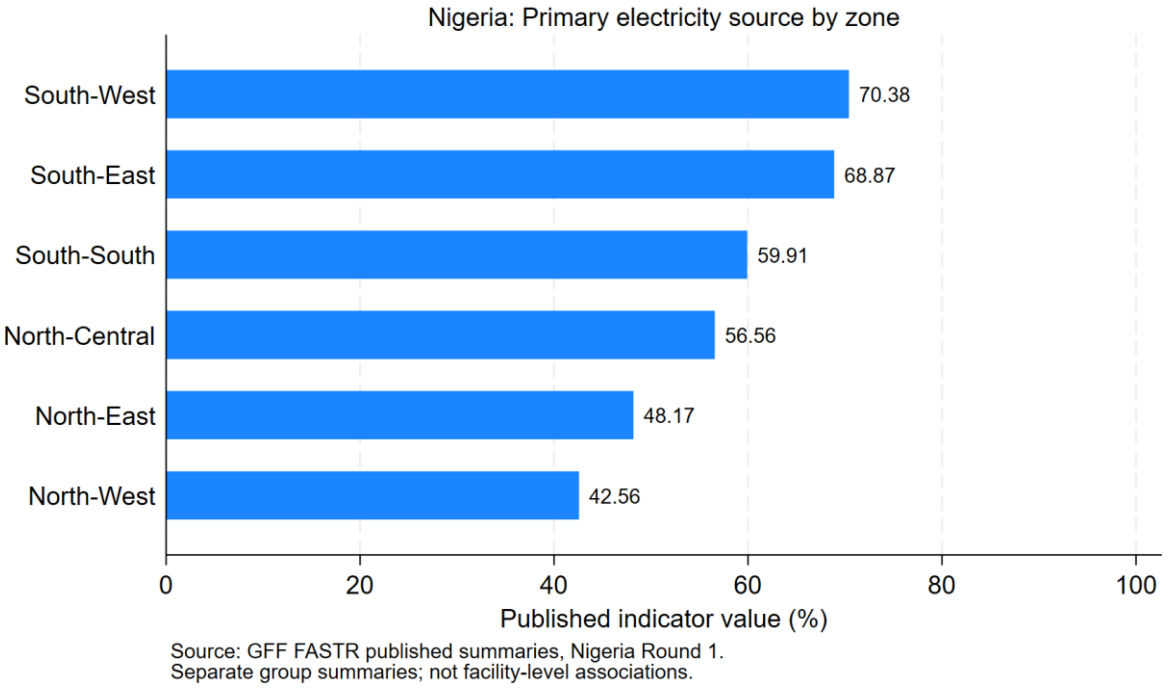


Figure A2. Facilities reporting a primary electricity source by zone.

Both charts use the same six zones, allowing the different rankings to be read alongside one another.

Appendix B Supporting programme charts

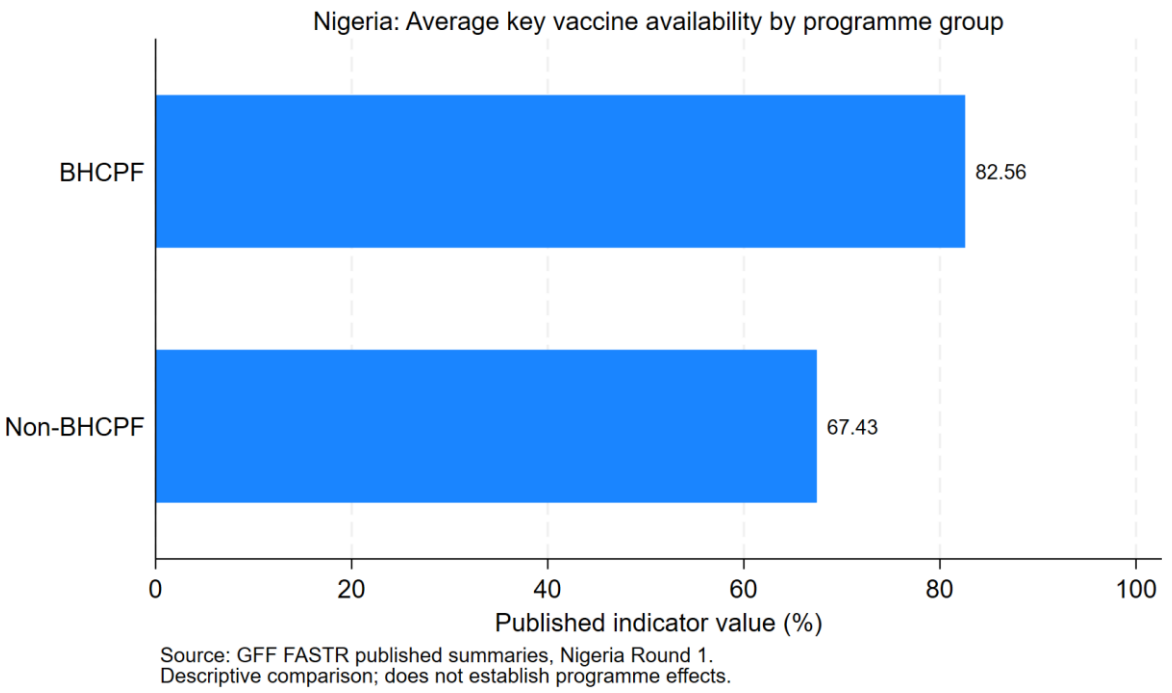


Figure B1. Published average key vaccine availability by programme group.

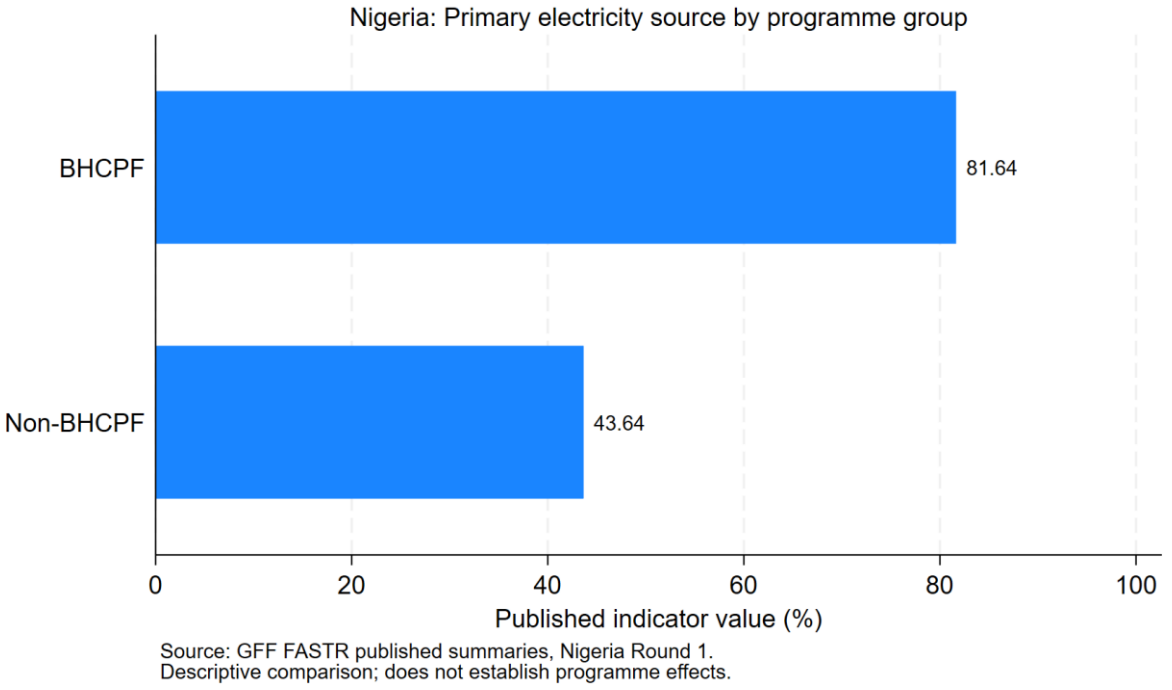


Figure B2. Facilities reporting a primary electricity source by programme group.

These are descriptive programme-group comparisons. BHCPF refers to the Basic Health Care Provision Fund.